

Kali Linux System Enumeration Report — Part 2

Labs 7-8 — Section 3 Practical Exercises

1. Introduction

This is the second part of the Section 3 practical report. It covers Lab 7 (network enumeration) and Lab 8 (an enumeration script). The labs were performed in the same Kali GNU/Linux Rolling 2026.2 VirtualBox virtual machine described in Part 1 (kernel 6.19.14+kali-amd64, XFCE, X11).

2. Lab 7 — Network Enumeration

Commands:

```
ip addr
ip -br addr
ip link
ip route
cat /etc/resolv.conf
hostname
hostname -I
sudo ss -tulpn
ip route | grep default
ping -c 4 127.0.0.1
```

Purpose: Identify the network interfaces and IP addresses (`ip addr` , `ip -br addr` , `hostname -I`), interface status (`ip link`), the routing table and default gateway (`ip route`), the DNS server (`/etc/resolv.conf`), listening ports (`ss`), and test that the local network stack works by pinging the loopback address (`ping -c 4 127.0.0.1`).

Result: No screenshots of this lab's output were provided, so its results (IP addresses, gateway, DNS, ping replies) are not documented here. The one earlier `ss -tulpn` run (Lab 4) showed no listening ports.

3. Lab 8 — Enumeration Script

Commands:

```
nano kali_enum.sh
chmod +x kali_enum.sh
./kali_enum.sh
```

What I did: Wrote a Bash script that runs the enumeration commands from the earlier labs in one go, printing a labeled heading before each section.

```
GNU nano 9.0      kalis.sh *
echo
echo "===== MEMORY ====="
free -h

echo
echo "===== DISK ====="
df -h

echo
echo "===== PCI DEVICES ====="
lspci

echo
echo "===== USB DEVICES ====="
lsusb

echo
echo "===== RUNNING SERVICES ====="
systemctl --type=service --state=running

echo
echo "===== "
echo "===== ENUMERATION DONE ====="
echo
```

Lab 8 evidence: nano editing the script (background chat window removed from the image)

What the screenshot shows: The script is open in GNU nano 9.0. Its last sections are `echo "===== MEMORY ====="` with `free -h`, DISK with `df -h`, PCI DEVICES with `lspci`, USB DEVICES with `lsusb`, RUNNING SERVICES with `systemctl --type=service --state=running`, and a closing "ENUMERATION DONE" banner. The title bar shows the file name `kalis.sh` with an asterisk, meaning it had unsaved changes. Note that this name differs from `kali_enum.sh` in the commands above.

Not visible: The top of the script, saving the file, `chmod +x`, and running the script. The screenshot does not show the script's output.

Concepts: Bash scripting (`#!/bin/bash`, `echo` headings), file editing with nano, making a file executable with `chmod +x`, and running a script with `./`.

4. Summary

Lab	Topic	Evidence provided
7	Network enumeration	Commands only; no output screenshots
8	Enumeration script	One screenshot of the script in nano

Skills practiced: checking interfaces, IP addresses, routes, DNS and listening ports with `ip`, `ss` and `/etc/resolv.conf`; testing connectivity with `ping`; writing and running a Bash script with `nano` and `chmod +x`.

Security relevance: Network details (addresses, routes, DNS, open ports) show how a system is reachable and what it exposes. Automating these checks in a script makes enumeration repeatable and consistent.

Unrelated personal content visible in the screenshot background was removed from the image.